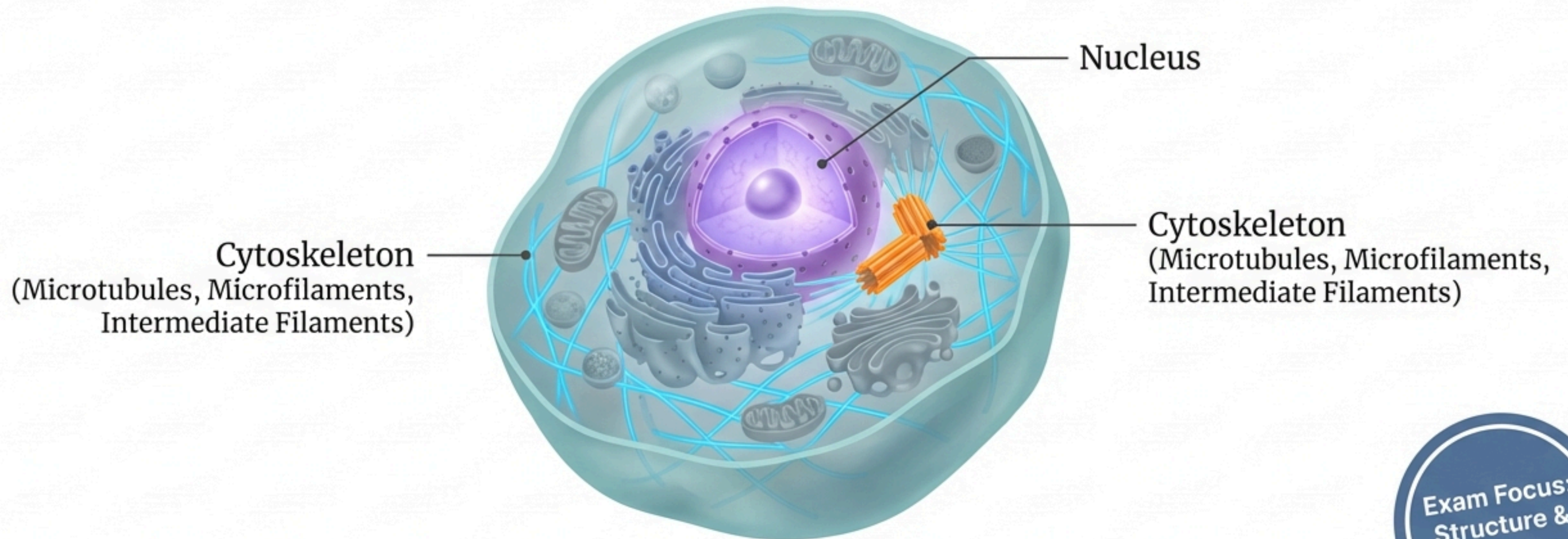


Architectural Foundations of the Eukaryotic Cell

High-Yield Revision: Cytoskeleton, Cilia, Centrosomes, & Nucleus

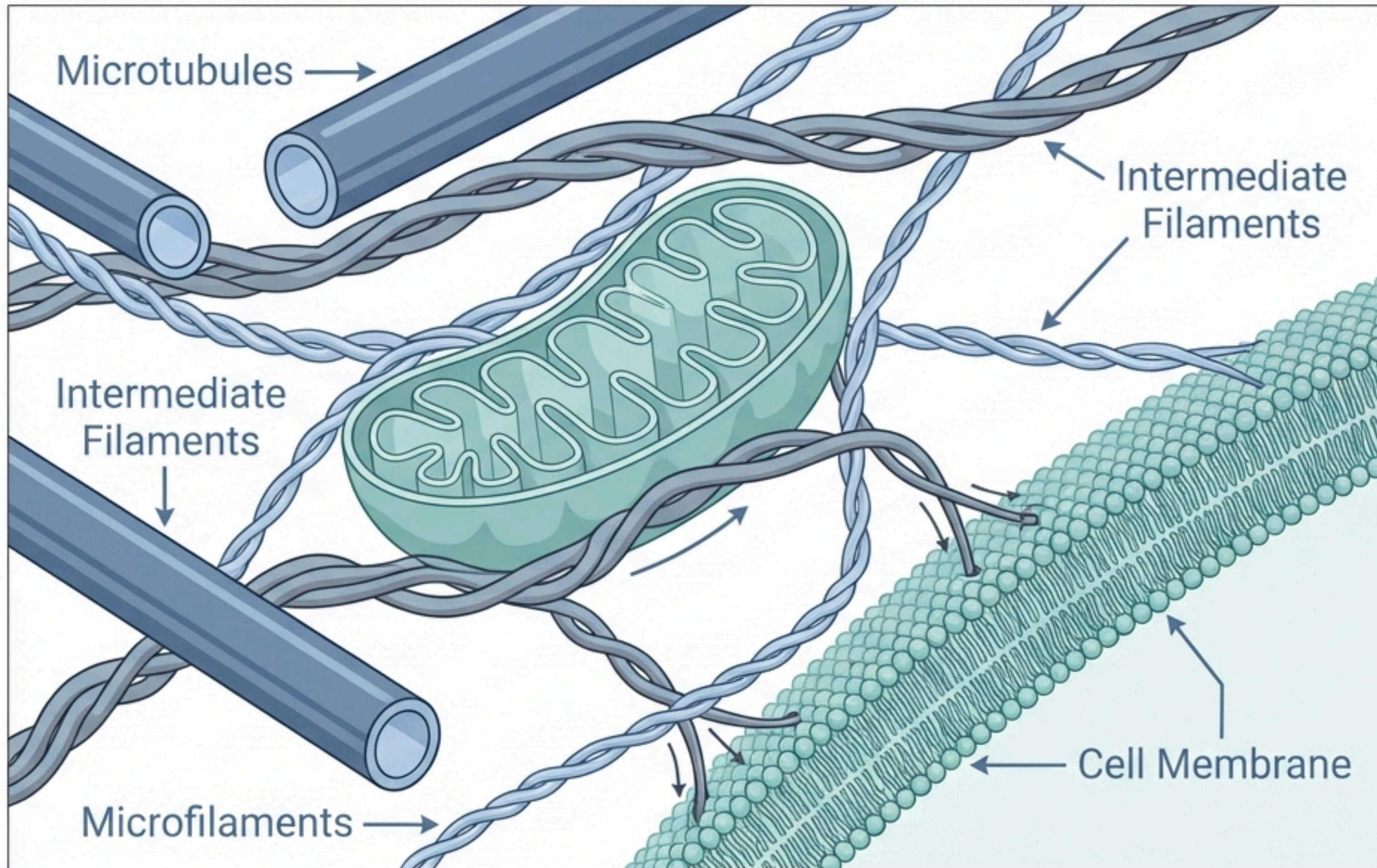


Based on Standard Exam Source Material

Exam Focus:
Structure &
Function

The Cytoskeleton: An Elaborate Network

Cytoplasmic Protein Meshwork (3D View)



Definition

An elaborate network of filamentous proteinaceous structures present in the cytoplasm.

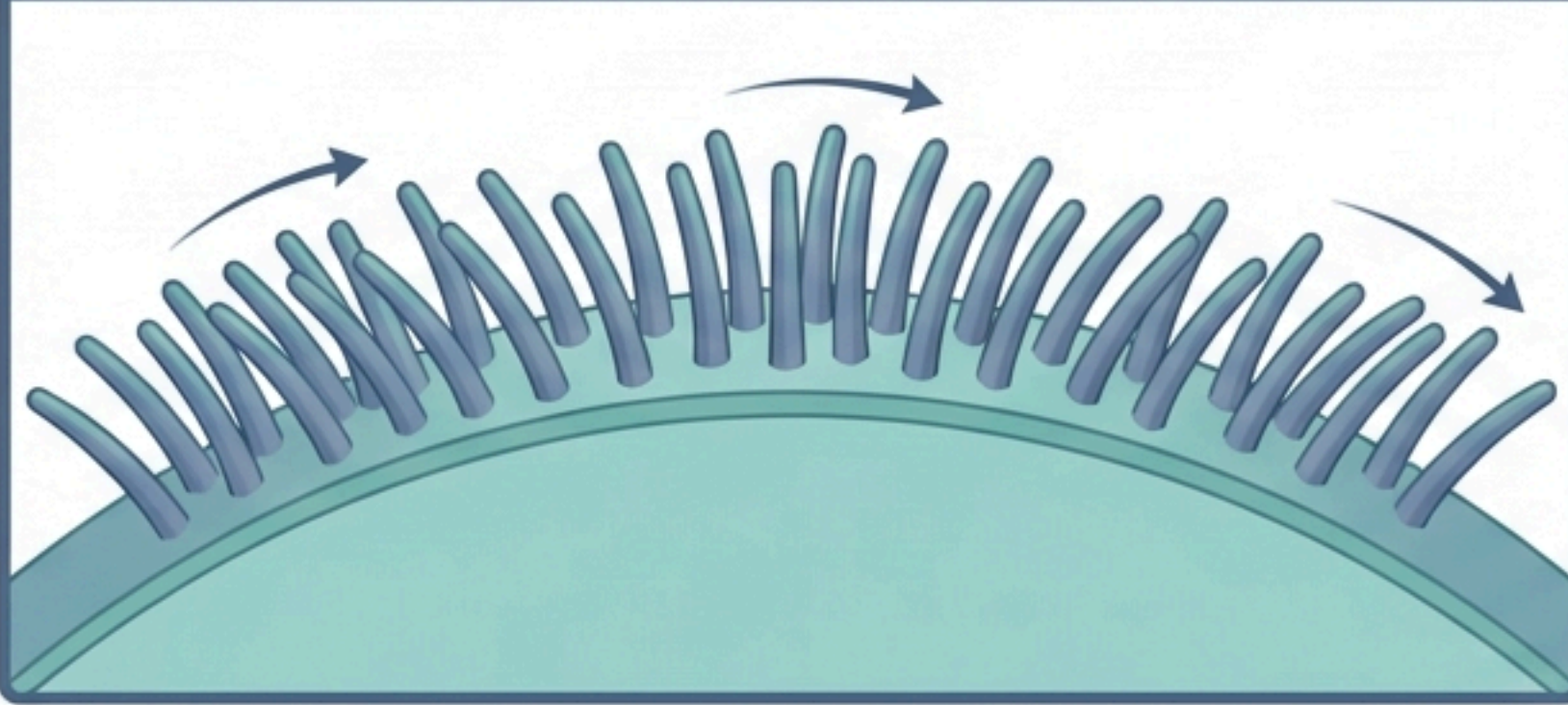
Collectively referred to as the cytoskeleton.

Key Functions

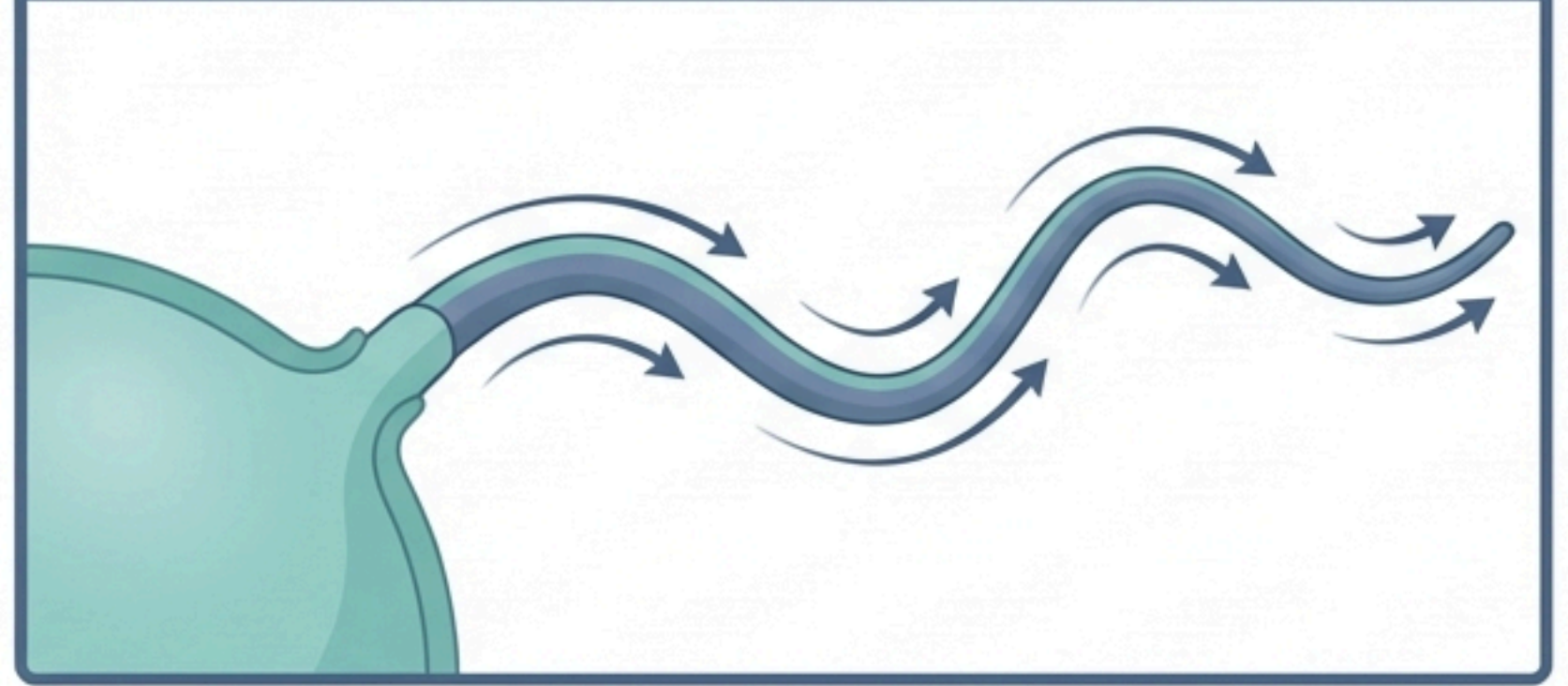
- Mechanical support
- Motility
- Maintenance of the shape of the cell

Cilia and Flagella: The Movers

Cilia (Singular: Cilium)



Flagella (Singular: Flagellum)

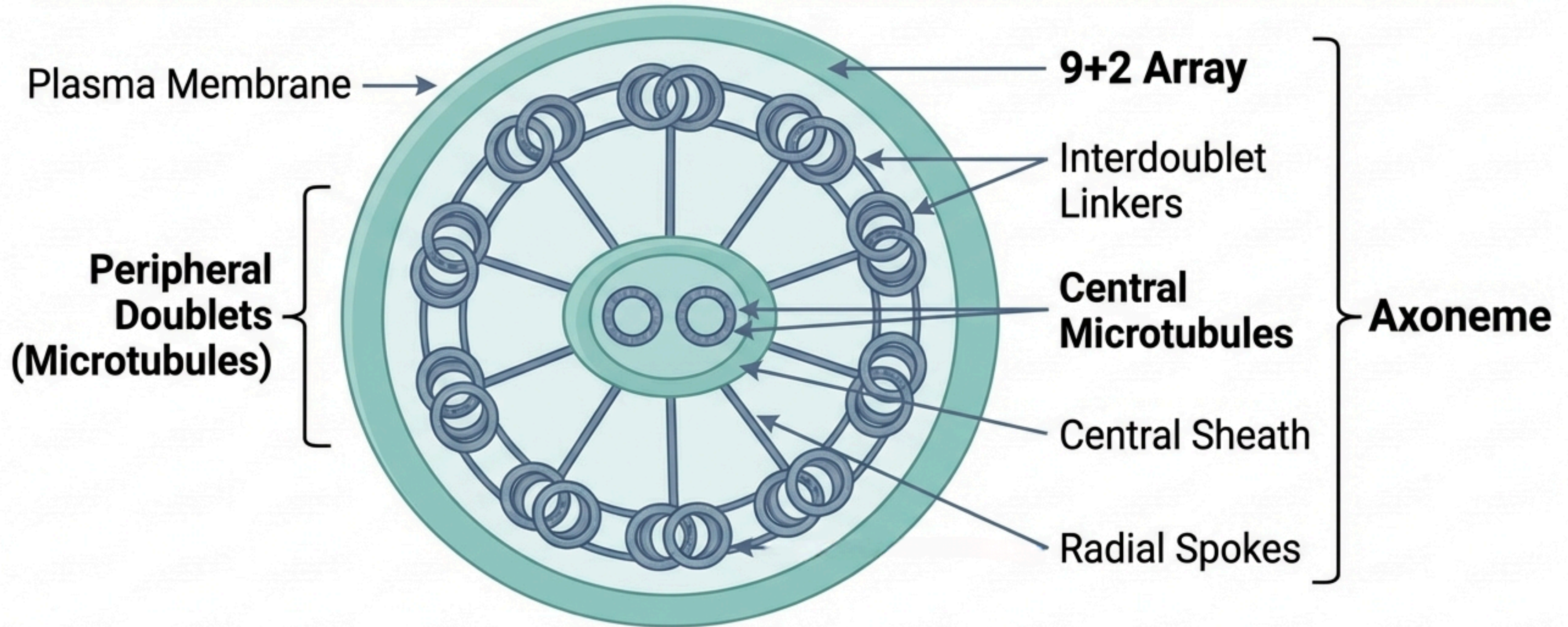


	Cilia	Flagella
Both are hair-like outgrowths of the cell membrane.	Small structures. Work like oars. Cause movement of either the cell or the surrounding fluid.	Comparatively longer. Responsible for cell movement.

Caution: The prokaryotic bacteria also possess flagella but these are structurally different from that of the eukaryotic flagella.

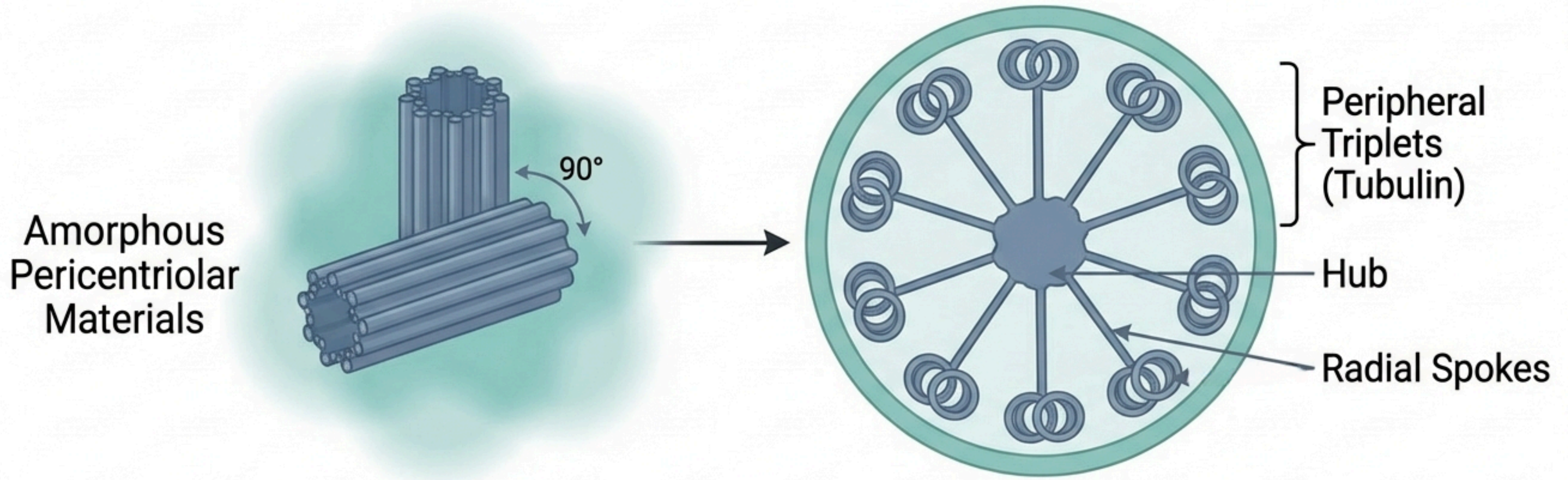
Anatomy of the Axoneme (9+2 Array)

Such an arrangement of axonemal microtubules is referred to as the 9+2 array.



Emerges from centriole-like structure called the basal bodies.

Centrosome and Centrioles: The Cartwheel Organization



Structure: Two cylindrical centrioles lying perpendicular to each other.

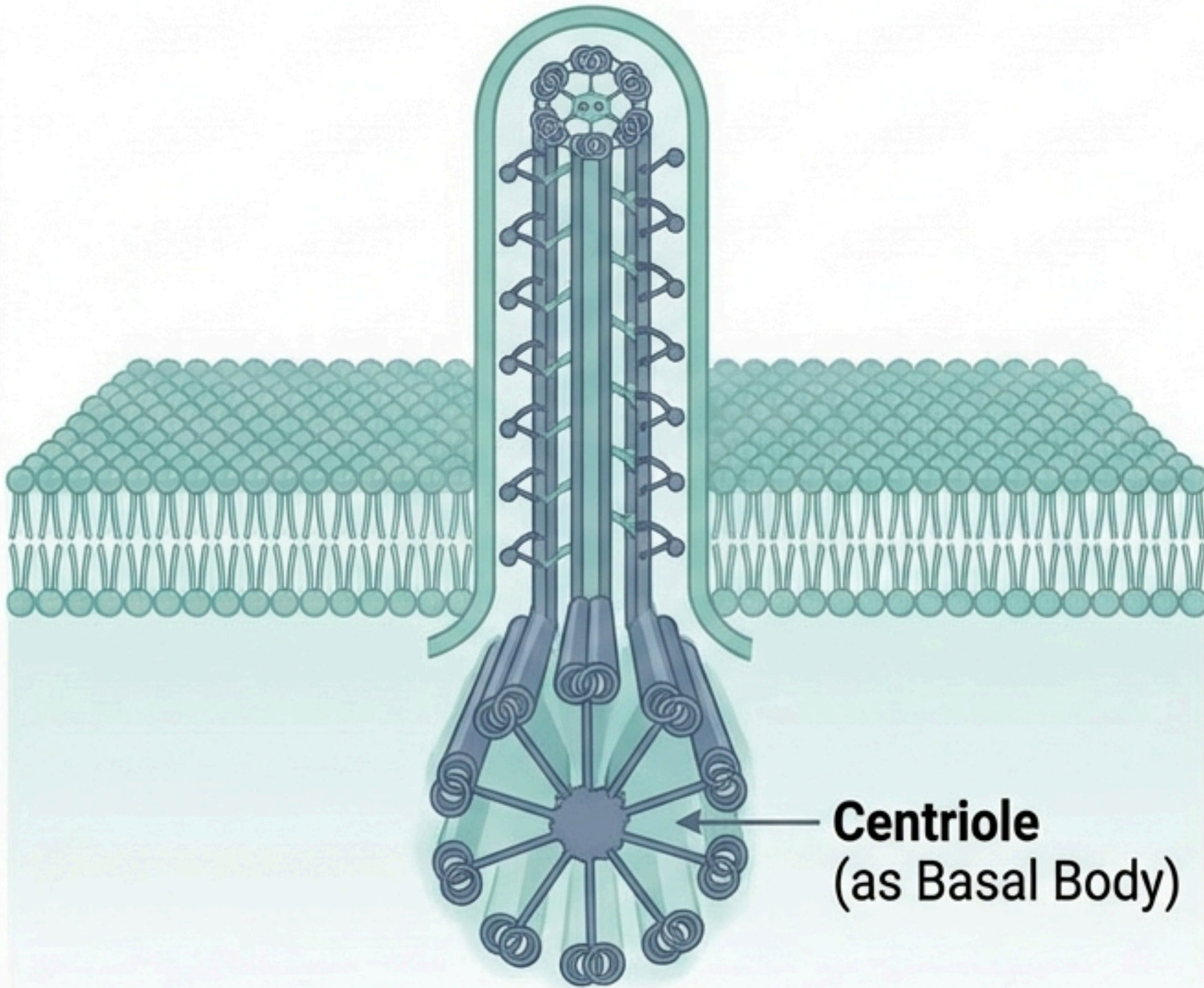
Composition: 9 evenly spaced peripheral fibrils of tubulin protein.

Specifics: Each peripheral fibril is a Triplet. Adjacent triplets are linked.

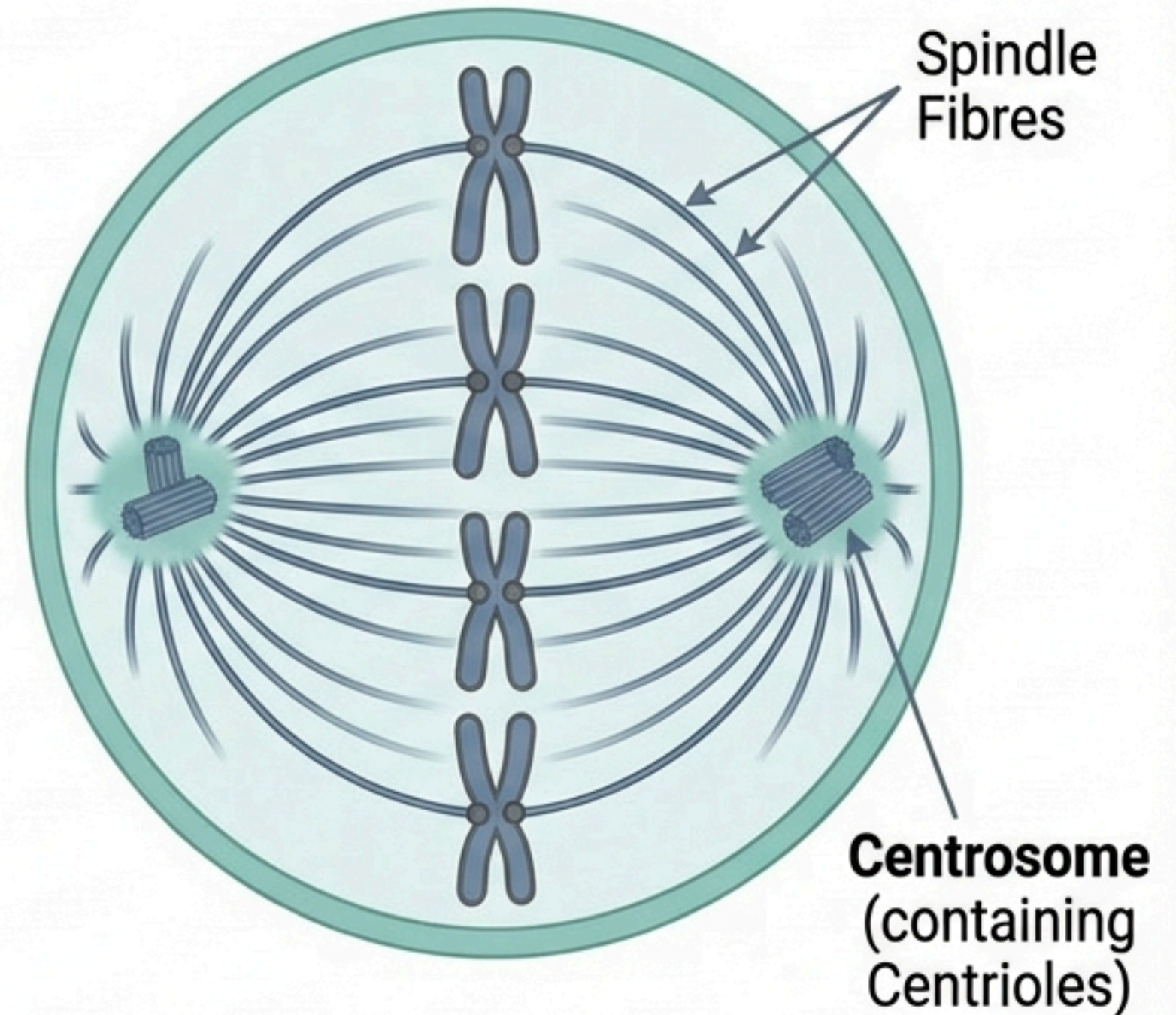
Each has an organisation like the cartwheel.

Functional Roles of Centrioles

Forms the Basal Body of Cilia and Flagella



Forms Spindle Fibres (Spindle Apparatus)



Specific to Animal Cells.

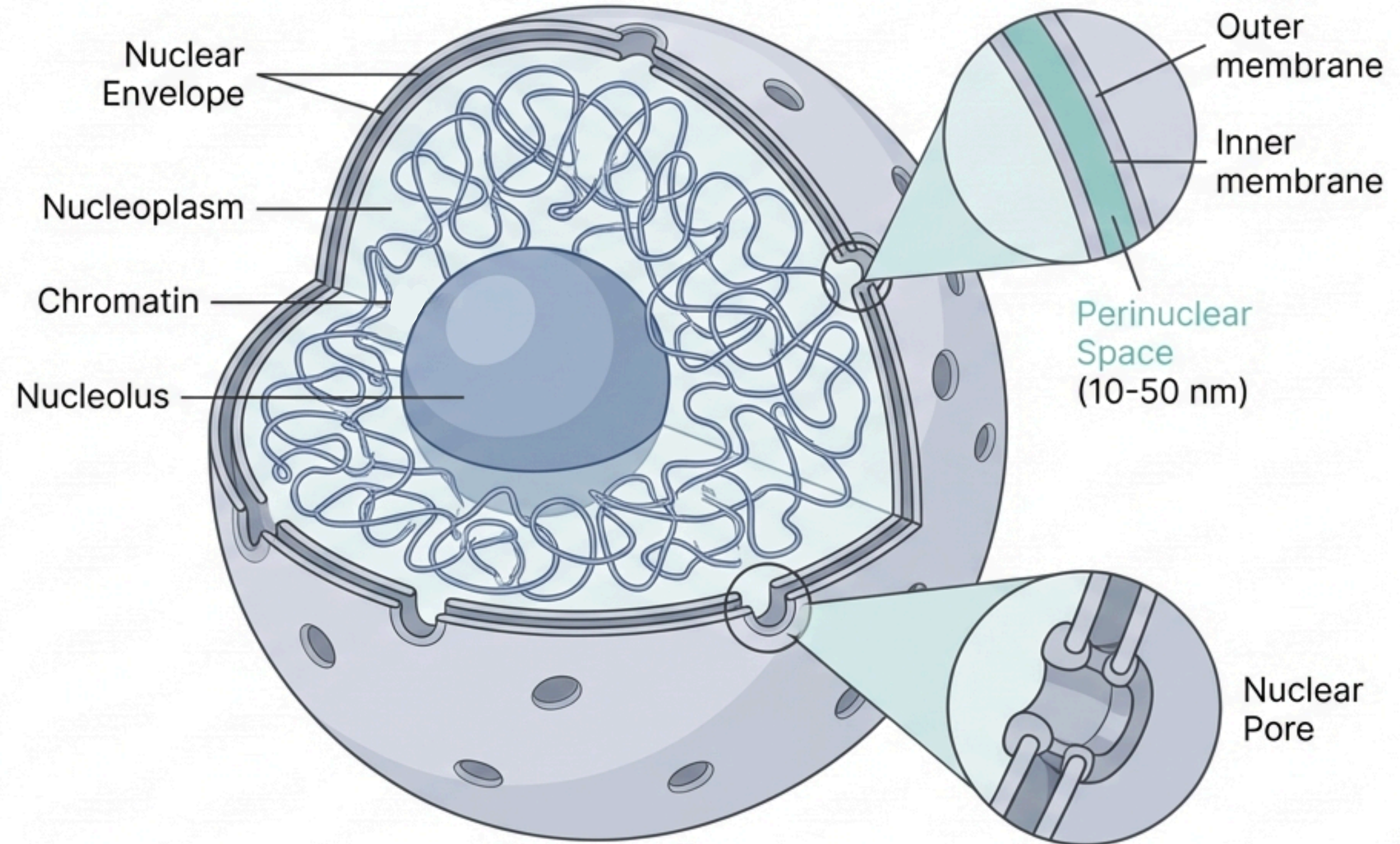
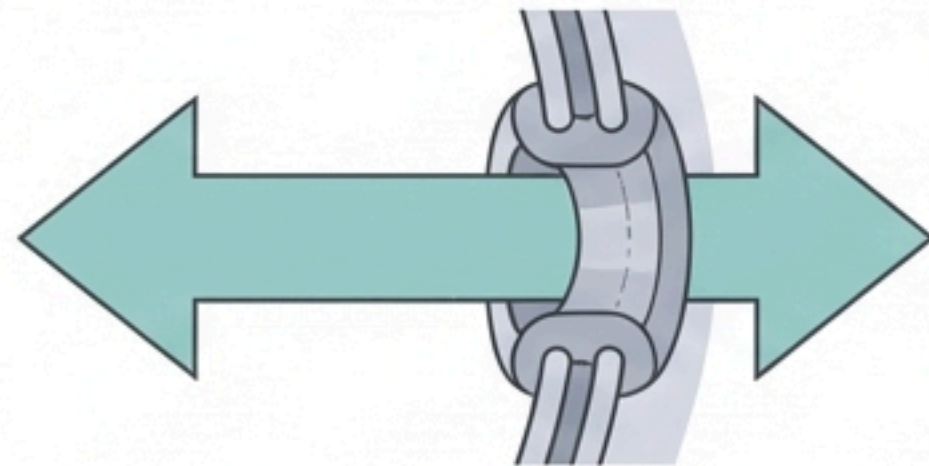
The Nucleus: Control and Transport

Discovery: Robert Brown (1831)

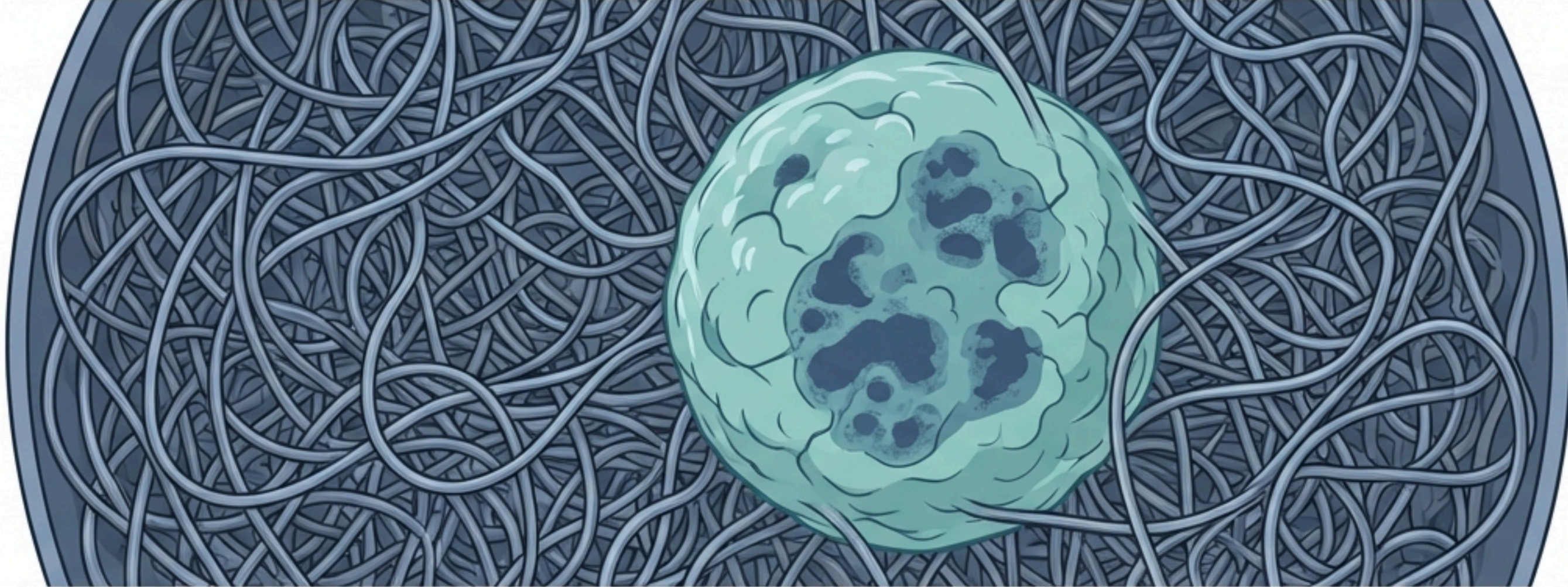
Naming: “Chromatin” named by Flemming (stained by basic dyes).

Envelope: Two parallel membranes. Outer membrane continuous with Endoplasmic Reticulum (ER).

Movement of RNA and Protein takes place in BOTH directions.



Inside the Matrix: Nucleoplasm & Nucleolus



Nucleoplasm

The nuclear matrix containing nucleolus and chromatin.

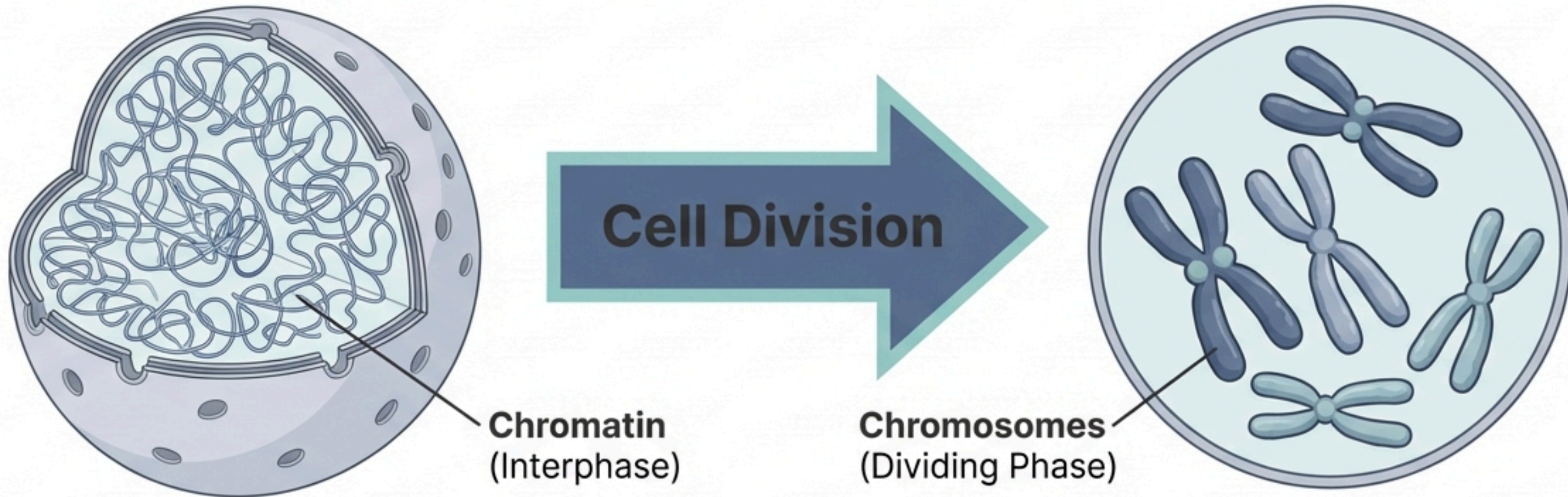
The Nucleolus

- Spherical structure.
- **NOT membrane-bound**: Content is continuous with nucleoplasm.
- **Function**: Site for active ribosomal RNA (rRNA) synthesis.

Insight:

Larger and more numerous nucleoli are present in cells actively carrying out protein synthesis.

Chromatin vs. Chromosomes



Chromatin Composition

- | | |
|---|--|
| ☒ DNA | ☒ Non-histone proteins |
| ☒ Basic proteins called Histones | ☒ RNA |

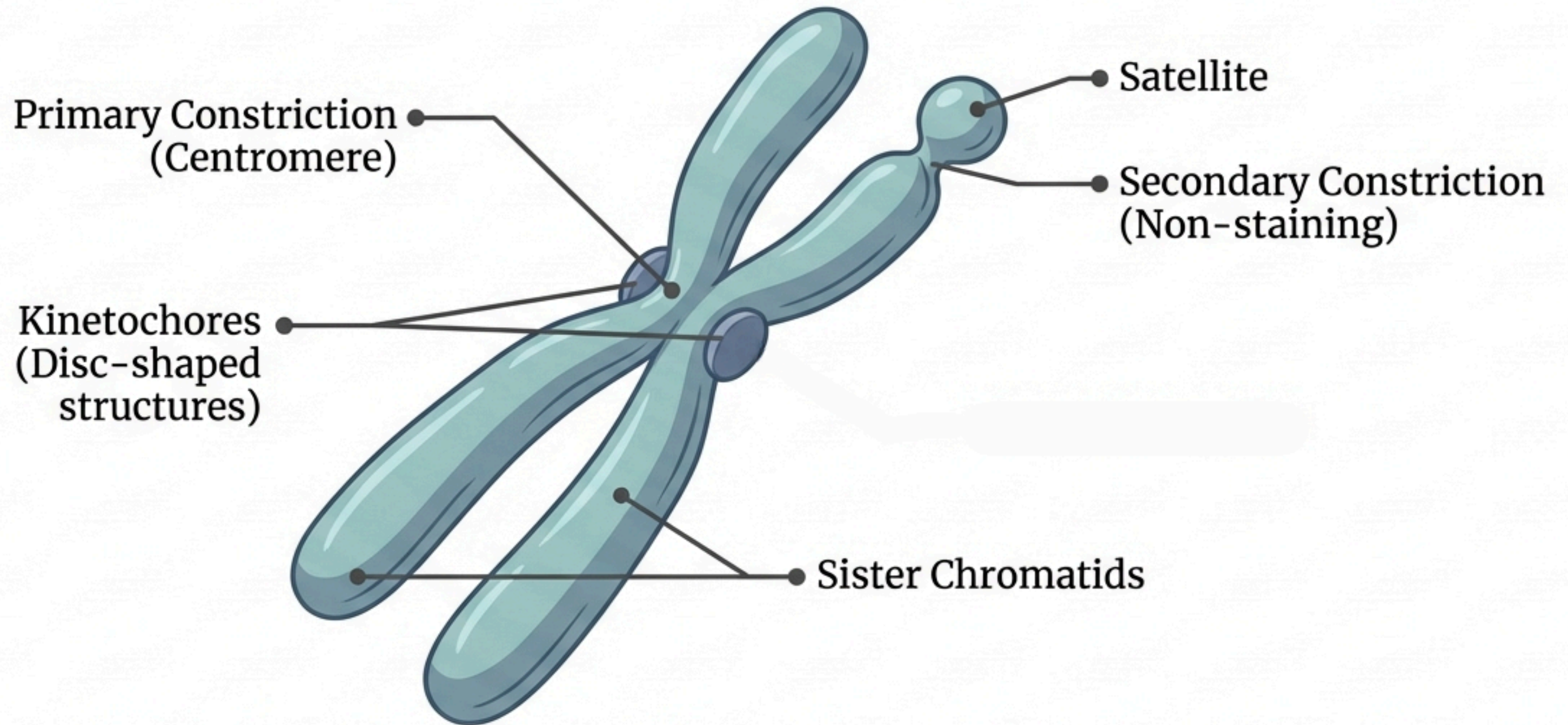
The Human Stat Box

A single human cell contains ~2 meter long thread of DNA distributed among 46 chromosomes (23 pairs).

Golden Line

Loose and indistinct network of nucleoprotein fibres called chromatin.

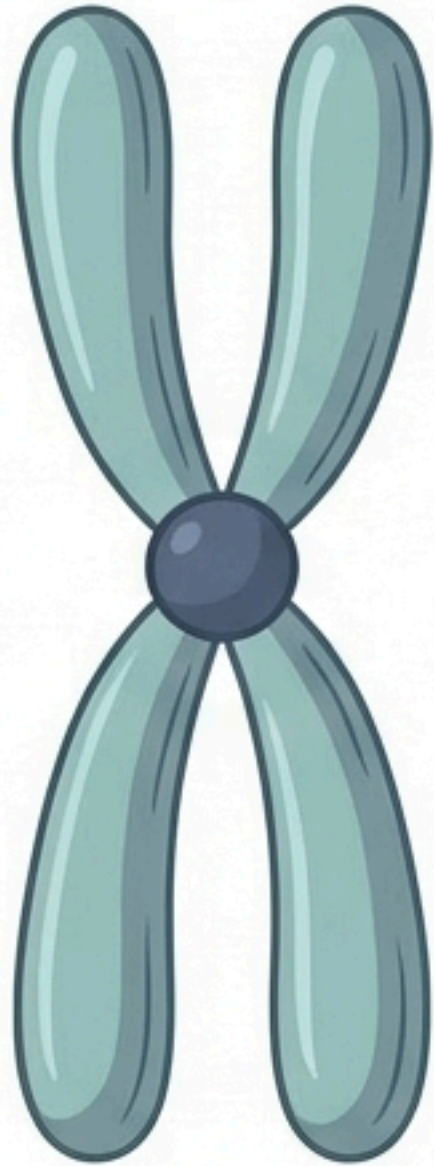
Anatomy of a Chromosome



Golden Line

Every chromosome essentially has a primary constriction or the centromere.

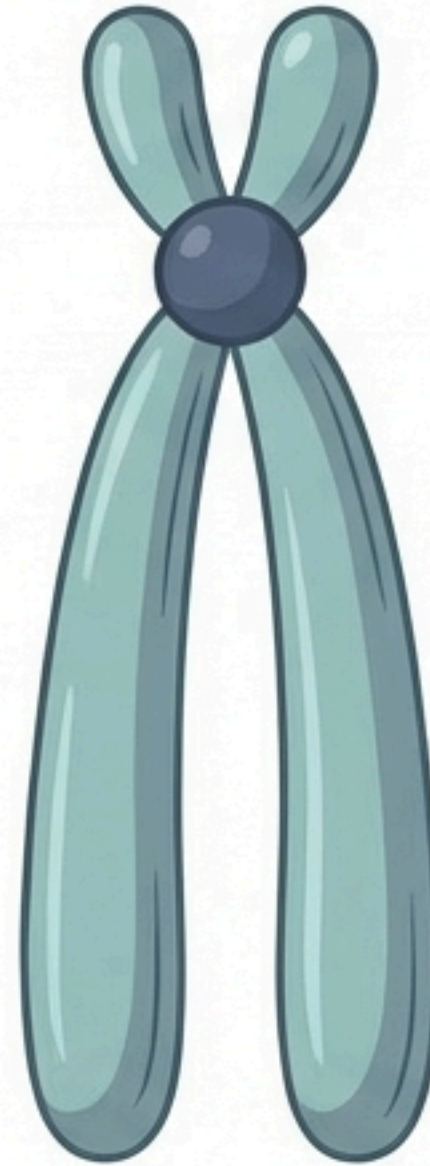
Classification Based on Centromere Position



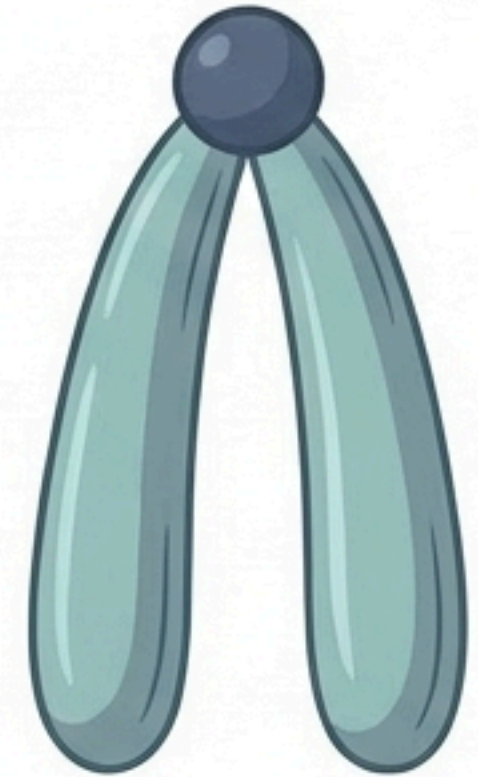
Metacentric
(Middle Centromere)



Sub-metacentric



Acrocentric



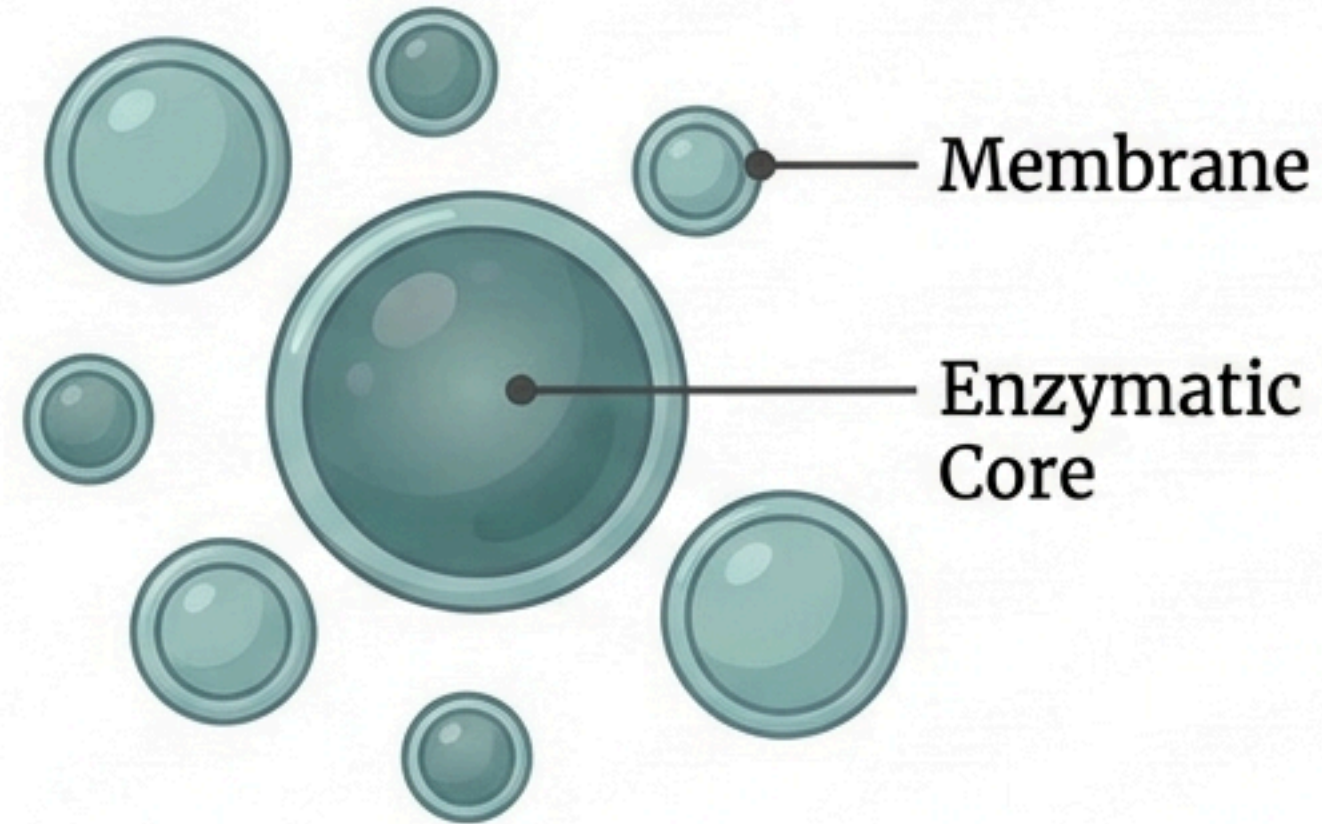
Telocentric
(Terminal Centromere)

Merriweather Regular

Classification is strictly based on the position of the centromere.

Microbodies & The 'Living' Cell Paradox

Microbodies



Microbodies: Membrane-bound minute vesicles containing various enzymes. Present in both plant and animal cells.

The Paradox

Exception Alert: Some mature cells lack a nucleus.



Erythrocytes
(Mammals)

Sieve Tube Cells
(Vascular Plants)



Merriweather Regular

Critical Thinking Question: Would you consider these cells as 'living'?

NEET-UG Challenge: Set 1 (Cytoskeleton & Cilia)

1. The cytoskeleton is made up of:
(A) Calcium carbonate structures (B) Proteinaceous filamentous structures
(C) Cellulosic fibres
(D) Phospholipid bilayers
2. In the '9+2' array of the axoneme, the radial spokes connect:
(A) Peripheral doublets to central sheath (B) Peripheral doublets to each other
(C) Central tubules to each other
(D) Basal body to plasma membrane
3. Which of the following statements regarding Cilia is TRUE?
(A) Structurally identical to bacterial flagella (B) Work like oars
(C) Have a 9+0 arrangement (D) Lack a plasma membrane
4. The arrangement of peripheral fibrils in a Centriole is:
(A) 9 Doublets (B) 9 Triplets (C) 9 Singlets (D) 9+2 Array
5. Centrioles form the basal body of cilia and which other structure in animal cells?
(A) Spindle fibres (B) Nucleolus (C) Microbodies (D) Kinetochores

NEET-UG Challenge: Set 2 (Nucleus & Chromosomes)

6. The perinuclear space measures: (A) 10–50 mm (B) 1–5 nm (C) 10–50 nm (D) 100–500 nm.
7. Larger and more numerous nucleoli are found in cells actively synthesizing: (A) Lipids (B) Proteins (C) DNA (D) Carbohydrates.
8. Which of the following is NOT a component of Chromatin? (A) DNA (B) Histones (C) RNA (D) Peptidoglycan.
9. A chromosome with a terminal centromere is classified as: (A) Metacentric (B) Telocentric (C) Acrocentric (D) Sub-metacentric.
10. The disc-shaped structures present on the sides of the centromere are called: (A) Satellites (B) Chromatids (C) Kinetochores (D) Secondary constrictions.

Answer Key & Summary

****Set 1****

- 1 - (B)
- 2 - (A)
- 3 - (B)
- 4 - (B)
- 5 - (A)

****Set 2****

- 6 - (C)
- 7 - (B)
- 8 - (D)
- 9 - (B)
- 10 - (C)

A typical eukaryotic cell is a complex factory. From the **Cytoskeleton** that maintains shape, to the **Centrioles** that organize division, and the **Nucleus** that holds the hereditary master plan.

“Thus, cell is the structural and functional unit of life.”